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Roll No

FT-701-CBGS
B.Tech., VII Semester
Examination, December 2020
Choice Based Grading System (CBGS)
Heat Transfer
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) State and explain Fourier's law for heat transfer. Mention the assumptions on which it is based. 7
b) What is Critical thickness? Why an insulated small diameter wire has a higher current carrying capacity than an uninsulated one? 7
2. a) Discuss the various regimes of nucleate boiling and explain the conditions for the growth of bubble. What is the effect of bubble size on boiling? 7
b) Explain the concept of hydrodynamic and thermal boundary layers. 7
3. a) Explain the following : Capacity ratio, Heat exchanger effectiveness, Number of transfer units. 7
b) What is thermal conductivity? How does it vary with the temperature vary in solid? 7
4. a) Write down the equation for conduction of heat through a slab or plane wall. 7

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- b) Write down the general equation for one dimensional steady state heat transfer in slab or plane wall with and without heat generation. 7
- 5. a) Define overall heat transfer co-efficient. 7
- b) Differentiate between conduction and convection. 7
- 6. a) How is natural convection different from forced convection? 7
- b) Define pool boiling in detail. 7
- 7. a) Differentiate between boiling and condensation, explain in detail. 7
- b) Explain the effect of boiling point elevation and hydrostatic head vapour recompression. 7
- 8. a) What are the methods used for determination of shape factor explain any one? 7
- b) Define thermal conductivity and give its unit. 7
